

Ihsane Foudal

Data Engineering & Software Engineering
| Final-Year Engineering Student

Khouribga , Maroc
+212 6 57 86 61 14
ihssanfoudal@email.com
Ihsane Foudal | LinkedIn
Ihsane Foudal | Github

PROFESSIONAL PROFILE

Fifth-year Engineering student in Computer Science and Data Engineering at ENSA Khouribga, seeking a final-year internship (PFE). Focused on **Data Engineering, Data Analysis and Business Intelligence**, I also have skills in **Machine Learning, Deep Learning and Software Development**. I am looking to apply my skills through innovative projects and contribute to the development of high-value solutions. Rigorous, curious and autonomous, I am motivated to grow in a stimulating professional environment.

EDUCATION

Engineering Degree – Computer Science and Data Engineering **2024 – 2027**

National School of Applied Sciences (ENSA), Khouribga
3rd year of the Engineering Program – Expected graduation in 2027

Integrated Preparatory Classes **2022 – 2024**

National School of Applied Sciences (ENSA), Khouribga

PROFESSIONAL EXPERIENCE

Data & Business Intelligence Intern **2026**

OCP Group – Khouribga
— Preparation, cleaning and analysis of more than **5 000 requests** using **Python (Pandas)**.
— Design of a **star-schema Data Warehouse** using **SQL Server** and creation of decision-support dashboards with **Power BI and DAX**.
— Beneficiary segmentation using **K-Prototypes** and development of a profile assignment model.

Software Development Intern **2025**

ONEE – Khouribga
— Design and development of a **3-tier** intranet application to digitize employee management and administrative requests.
— Development using **PHP, MySQL, HTML, CSS and JavaScript**, integrating PDF document generation and dashboards with **Chart.js**.

ACADEMIC PROJECTS

Sentiment Analysis in Moroccan Darija – NLP Pipeline **2026**
— Collection, cleaning and processing of a dataset containing more than **83 000 YouTube comments** in Moroccan Darija.
— Comparison of **Machine Learning and Deep Learning** models using different vectorization techniques for **sentiment classification : positive, negative and neutral**.
— Best performance achieved through fine-tuning of **BERTDarija**, with experiment tracking using **MLflow and Weights & Biases (W&B)**.

Business Intelligence & Data Analysis **2025**
— Design of a decision-support solution for the analysis of medicines, their effectiveness and adverse effects.
— Implementation of a **Data Warehouse using SQL Server**, automation of **ETL processes with SSIS**, construction of an **OLAP Cube with SSAS**, and creation of **Power BI** dashboards.

Face Mask Detection – Computer Vision **2026**
— Development of a real-time face mask detection system for **images, videos and webcam streams**.
— Use of **YOLO and VGG** models for detection and classification, with frame-by-frame processing and visualization of **confidence scores**.

Web Platform for Sales Management & Analysis **2026**
— Development of a platform for managing products, sales and inventory using **Java, Spring Boot, Thymeleaf and MySQL**.
— Implementation of an **MVC** architecture, **REST APIs**, and role-based authentication and authorization using **Spring Security**.
— Development of responsive interfaces using **HTML, CSS, Bootstrap and JavaScript**, integrating dashboards and sales analysis indicators.

TECHNICAL SKILLS

— **Data & Business Intelligence :** Python (Pandas, NumPy, Matplotlib, Scikit-learn, PyTorch), SQL, Excel, Power BI, DAX, ETL, SSIS, SSAS.
— **Databases :** SQL Server, MySQL, MongoDB.
— **Artificial Intelligence :** Machine Learning, Deep Learning, NLP, Computer Vision.
— **Software Development :** Java, Spring Boot, Thymeleaf, REST APIs, JavaScript, Next.js, NestJS, PHP, HTML, CSS, Bootstrap.
— **Design & Modeling :** UML, Merise, relational and dimensional modeling.

SOFT SKILLS & LANGUAGES

— Analytical thinking and problem-solving.
— Teamwork and communication.
— Autonomy, adaptability and curiosity.

Arabic : Native
French : Fluent
English : Intermediate